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Adaptive Anomaly Detection Algorithm for Real-Time Identification of Phishing and Credential Stuffing Attacks in E-Learning Systems Using Deep Reinforcement Learning

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Abstract

The quick embrace of e-learning software programs and cloud-based LMS has greatly enhanced digital education through better access and scalability of the platform. Nevertheless, with the reliance on the use of online authentication platforms, there is also a higher susceptibility to cyberattacks such as phishing attacks and credential stuffing attacks. These attacks utilize vulnerabilities in the user authentication process, stolen credentials, and deceptive communication tactics to gain unauthorized access, steal academic credentials, and disrupt the entire learning process. Traditional methods of intrusion detection and static machine learning have proven less effective in adapting to dynamic attack patterns and have a high percentage of false positives. To mitigate these problems, this paper proposes a novel Adaptive Anomaly Detection Algorithm that uses Deep Reinforcement Learning (DRL) to Real-Time Identify Phishing and Credential Stuffing Attacks in E-Learning Systems. The proposed approach involves the use of behavioral analysis, sequential anomaly detection, and adaptive policy optimization. Data pre-processing, behavioral

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feature extraction, state-action-based deep reinforcement learning, attack categorization, and adaptive mechanisms will form the basis of the proposed algorithm. Among behavioral features considered are failed login attempts, rate of IP change, session time, URL reputation, and device mismatch. Evaluation through experiments was carried out using a mixed cybersecurity data set that contained 125,000 instances with phishing attacks, credential stuffing, and legitimate users' actions. The developed DRL model attained outstanding results with an accuracy of 98.4%, precision of 98.1%, recall of 97.8%, and F1-score of 97.9% while having a minimum of false positives of 1.9%. A comparative study indicates remarkable improvements compared to SVM, Random Forest, CNN, and LSTM techniques. The findings prove the effectiveness of adaptive reinforcement learning in improving cybersecurity capabilities in contemporary e-learning environments.

Keywords: Deep Reinforcement Learning, Adaptive Anomaly Detection, Phishing Attack Detection, Credential Stuffing Attacks, E-Learning Cybersecurity, Learning Management Systems, Real-Time Intrusion Detection.

1 Introduction

However, with the evolution and development of e-learning technologies, there has been a transformation in the education domain because of the presence of virtual classes, cloud LMS, online exams, and collaboration in academics (Nankya et al., 2024). E-learning solutions rely on web applications and remote authentication processes, among other cloud-based features, that make it possible for large-scale academic activities. In terms of increased security concerns, there are phishing attacks and credential stuffing that benefit from insecure authentication mechanisms, password reuse, and human behavior patterns (Bukhari et al., 2023). These lead to data theft that consists of sensitive information such as the students' data, exam scores, financial information, intellectual property, and the reputation of the organization (Kola et al., 2025).

The attacks through phishing usually consist of emails, fraudulent LMS login pages, and misleading communication links used to harvest victims' credentials (Freire, 2025). Another type of attack, known as credential stuffing, uses automated bots for testing the authenticity of usernames and passwords on various educational platforms. The problem with traditional intrusion detection systems and signature-based approaches is that, since modern-day attacks continuously change patterns and timing, the systems will not be able to recognize these changes and detect such attacks. In addition, a traditional approach leads to high levels of false positives and a lack of responsiveness to real-time anomalies that occur in distributed networks.

In order to address the stated problem, this study offer a framework that includes an Adaptive Anomaly Detection Algorithm for Real-Time Identification of Phishing and Credential Stuffing Attacks in E-Learning Systems Using Deep Reinforcement Learning (DRL) (Wei & Maidanov, 2026; Ramya, 2025). Such a system utilizes dynamic learning of network behavior, real-time anomaly identification, and optimized policies in order to detect cyber-attacks (Smith, 2025). The use of deep reinforcement learning allows the algorithm to continuously adjust its actions based on network behavior.

Key Contributions

- Offers an adaptive anomaly detection architecture using deep reinforcement learning for e-learning cybersecurity.
- Designs a dynamic method for detecting phishing and credential stuffing attacks in decentralized LMS systems.

- Incorporates behavioral analysis and recognition of sequential attack patterns to enhance anomaly detection performance.
- Lowers the false positive rate with policy adaptation based on continuous environmental feedback.
- Contributes to better cybersecurity resilience, trust, and secure user authentication management within the e-learning paradigm.

This paper is further structured as follows. Section II includes the literature review concerning cybersecurity, phishing attacks, and anomaly detection approaches in the context of e-learning environments. Section III describes the novel deep reinforcement learning-based approach for adaptive anomaly detection. Section IV addresses the experimental study conducted for evaluating the proposed system, including its setup, dataset characteristics, measures used, and comparative results analysis. Conclusions can be found in Section V.

2 Literature Survey

The rapid growth of online learning platforms has raised considerable cybersecurity risks in e-learning platforms, including phishing, credential theft, hacking, and automated cyberattacks. Research has been done on the use of artificial intelligence, machine learning, and adaptive security architecture to enhance threat detection in education systems (Abdullah, 2024) (Prasath, 2025).

The preceding article conducted an extensive survey on deep learning and computer vision techniques for cybersecurity in smart education (Ali et al., 2025; Hassan et al., 2024). It stressed the need to adopt AI-based intrusion detection systems that would ensure the protection of information and online learning platforms from cyber-attacks (Muzaffar, 2024). Also, an intrusion detection system was designed that could detect various types of cyber-attacks in AI-based online learning platforms (Ayoubi et al., 2025; Hasan & Marisna, 2024).

This research was related to adaptive cybersecurity strategies used in higher educational institutions and found that intelligent cybersecurity solutions help enhance the robustness of their digital infrastructure (Daoud & Hamdi, 2025; Ray, 2026; Gowsikraja, 2025). This research addressed the advanced detection of DDoS attacks in online learning platforms by analyzing network traffic intelligence using intelligent cybersecurity techniques (Berqia et al., 2025; Rawindaran et al., 2021; Ayoubi et al., 2018).

Keystroke dynamics and behavioral biometrics were studied in this research by making use of various machine learning and deep learning algorithms for user authentication (Gündoğan & Celiktas, 2025). According to their results, behavior analytics can detect unusual user activities and prevent the exploitation of stolen credentials (Rawindaran et al., 2021; Samout et al., 2025). The use of password less authentication was also addressed in the paper (Yusop et al., 2025).

The research used secure e-learning architecture in association with data mining and Industry 4.0 for showcasing the intelligence of analytics as a mechanism for improving cyber defense (Vinita & Reshu, 2025). Another important contribution from the literature is the proposal of the use of CNN for ontology-based security prediction, which shows the benefits of deep learning in the context of anomaly detection (Devi & Deepa, 2024; Soy & Salwadkar, 2024).

While the current state of research contributes greatly to the domain of AI-based cybersecurity and intrusion detection, most of the approaches in the area involve the use of supervised learning or

classification models that do not have any adaptive learning properties when facing evolving attack patterns (Anjum et al., 2025; Oyekan & Enyejo, 2023). Moreover, there is a limited body of literature focusing on the combined detection of phishing and credential stuffing attacks within e-learning environments using a reinforcement learning approach.

Accordingly, the aim of the suggested research project is to implement a Deep Reinforcement Learning-based Adaptive Anomaly Detection Framework, which will be able to adaptively learn the behavior of the attacks, continuously adjust detection mechanisms, and thus improve threat detection accuracy within modern e-learning platforms.

3 Proposed Model

3.1 Overall Methodology

The proposed system introduces an Adaptive Anomaly Detection Framework based on Deep Reinforcement Learning (DRL) to detect any signs of phishing and credential stuffing in the e-learning system at real-time intervals. It is designed to monitor any authentication process, user network behavior, request sequences, and session properties in Learning Management Systems (LMS). In contrast to previous static models, the new system is capable of learning different patterns of attacks and adapting the decision-making policy according to the changing nature of attacks.

There are five main phases of the methodology; these include data collection, data preprocessing, feature selection, anomaly detection using deep reinforcement learning, and the generation of adaptive responses. At first, login requests, IP addresses, session times, unsuccessful logins, browser fingerprints, location attributes, and URL attributes are gathered from the environment of the e-learning application. Next, this gathered data is preprocessed and converted into feature vectors.

Following the preprocessing stage, behavioral and temporal features are generated to detect any suspicious behaviors, which include multiple unsuccessful login attempts, unusual frequency of accessing the website, switching of devices that are not authorized, and clicking on phishing URLs. Such features are used by the DRL agent for detecting any attacks in the online network environment.

In such a way, the process involves monitoring of system states, performing security actions, receiving reward signals from the system states, and updating the detection policy based on the obtained results. When the agent is able to make correct attacks, positive rewards are given, whereas any wrong attacks result in a penalty. In such a way, the proposed approach is capable of recognizing evolving phishing attacks and automated credential stuffing attacks.

Eventually, after the anomalies have been detected, appropriate adaptive security measures will be generated.

Figure 1 depicts the architecture of the suggested approach for cybersecurity using deep reinforcement learning on e-learning platforms. It shows the steps involved from collecting and extracting data from e-learning systems, to performing anomaly detection and classification of attacks, to providing automatic responses to mitigate these attacks.

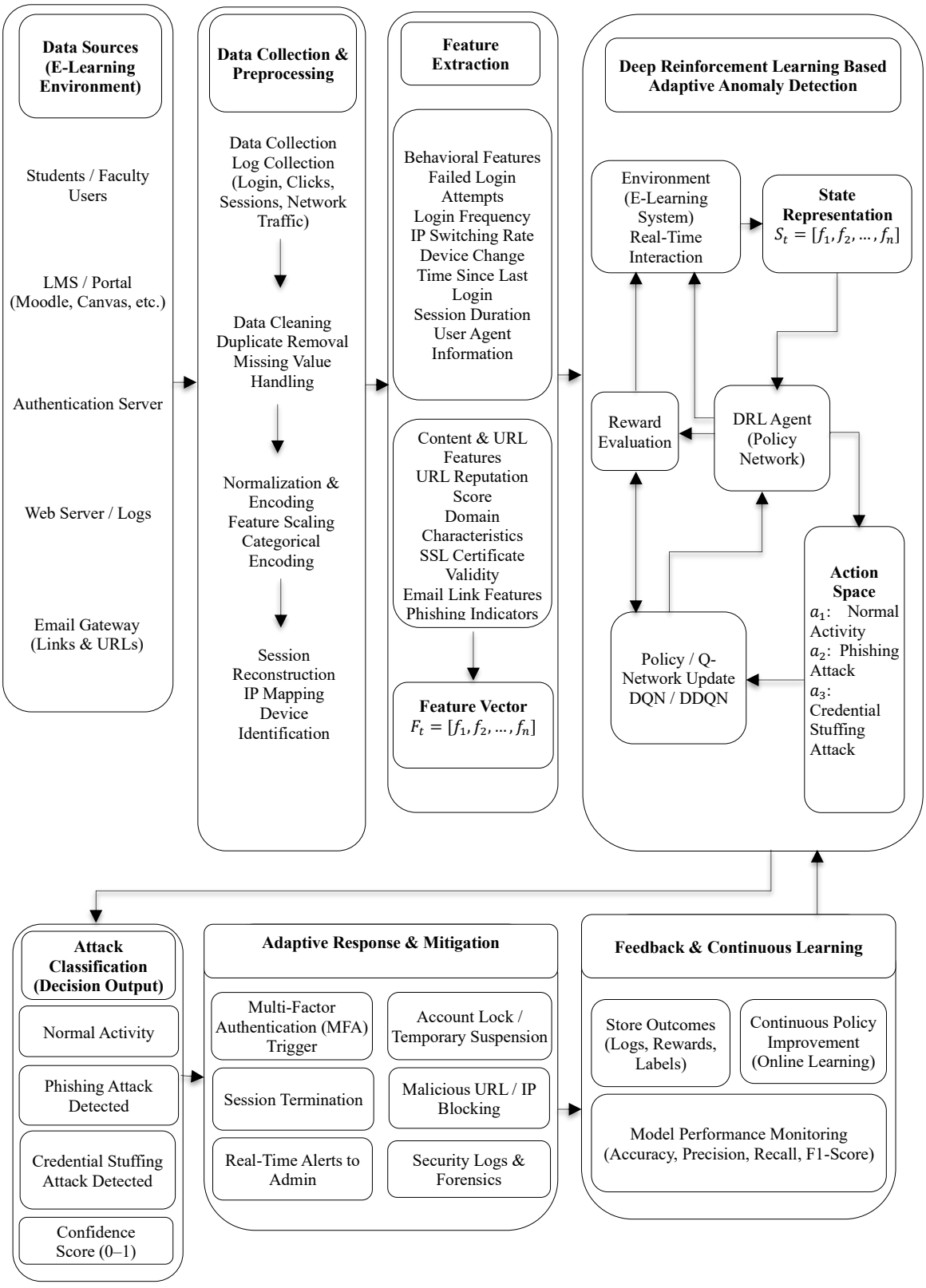


Figure 1: Architecture of adaptive anomaly detection framework for real-time identification of phishing and credential stuffing attacks in e-learning systems using deep reinforcement learning

3.2 Deep Reinforcement Learning Algorithm

Algorithm 1: Adaptive DRL-Based Anomaly Detection

Input: User login activities L, Session behavior features S, Network traffic data N

Output: Attack classification result, Adaptive mitigation response

Steps:

1. Initialize DRL environment and detection policy.
2. Collect real-time LMS authentication and session data.
3. Preprocess and normalize input data.
4. Extract behavioral anomaly features.
5. Observe the current system state s_t.
6. DRL agent selects action a_t:
   - Normal
   - Phishing attack
   - Credential stuffing attack
7. Execute selected action and evaluate outcome.
8. Compute reward r_t:
   - Positive reward for correct detection
   - Negative reward for false classification
9. Update Q-network/policy parameters.
10. Repeat learning until optimal detection accuracy is achieved.
11. Trigger adaptive security response for detected attacks.

The proposed Deep Reinforcement Learning-Based Adaptive Anomaly Detection Algorithm (Algorithm 1) keeps track of the actions performed by users while logging in, their activities on the network, and session behaviors in order to detect phishing and credential-stuffing attacks in real time. It works by improving its policies based on the environmental feedback it receives through reward-based optimization.

3.3 Mathematical Description

Feature Vector Representation

The extracted behavioral features are represented as equation 1:

$$\mathrm{F} = \{\mathrm{f}_1, \mathrm{f}_2, \mathrm{f}_3, \ldots, \mathrm{f}_\mathrm{n}\} \quad (1)$$

Where, f_1= Failed login attempts, f_2= Login frequency, f_3= Session duration, f_4= IP switching rate, f_5= URL reputation score, f_n= Additional behavioral attributes.

State Representation

The system state at time t is defined as equation 2:

$$s_t = [f_1, f_2, f_3, \ldots, f_n] \quad (2)$$

The DRL agent continuously observes the current environment state for anomaly detection.

Action Space

The action set is represented as equation 3:

$$A = \{a_1, a_2, a_3\} \quad (3)$$

Where, a_1= Normal activity, a_2= Phishing attack, a_3= Credential stuffing attack.

Reward Function

The reward mechanism guides adaptive learning is shown in equation 4:

$$R(s_t, a_t) = \begin{cases} +1, & \text{Correct detection} \\ -1, & \text{False positive} \\ -2, & \text{Missed attack} \end{cases} \quad (4)$$

This reward structure improves attack classification efficiency over time.

Q-Value Update Equation

The DRL model updates its policy using the Q-learning optimization rule, which is represented as equation 5:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_t + \gamma \max Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (5)$$

Where, α= Learning rate, γ= Discount factor, r_t= Reward at time t, $Q(s_t, a_t)$= Current Q-value

Attack Detection Probability

The final anomaly classification probability is calculated using Softmax activation, as shown in equation 6:

$$P(a_i \mid s_t) = \frac{e^{Q(s_t, a_i)}}{\sum_{j=1}^{n} e^{Q(s_t, a_j)}} \quad (6)$$

The chosen class is the one that has the maximum probability.

The approach offers adaptive, intelligent, and scalable cybersecurity defense for current e-learning systems by integrating behavioral analysis with real-time anomaly detection through deep reinforcement learning.

4 Results and Discussion

4.1 Software and Implementation Details

The presented adaptive anomaly detection algorithm was implemented in Python 3.11 with the use of machine/deep learning algorithms and libraries for cybersecurity analysis tasks. The experiments were

run on an Ubuntu Linux machine with an Intel Core i7 processor, 32 GB RAM and NVIDIA RTX graphics card. For the construction of the Deep Reinforcement Learning (DRL) model, TensorFlow and Keras libraries were used. The Scikit-learn library was utilized for performing data preprocessing, feature engineering, and evaluation of the algorithm performance. Moreover, Pandas and NumPy libraries were utilized for conducting cybersecurity analysis of LMS log data and network traffic. The visualization of results was made possible by means of the Matplotlib library. For the experiments, there was used a mixture cybersecurity dataset consisting of 125,000 records, including 72,000 normal ones, 31,000 related to phishing, and 22,000 representing cases of credential stuffing. The mixture cybersecurity dataset contains 24 numeric and categorical features with a split into 70% of training, 15% of validation, and 15% of testing datasets. The major hyperparameters of the DRL are as follows: learning rate is equal to 0.001, discount factor is equal to 0.95, batch size equals 64, replay memory size equals 10,000, and the number of trainings.

4.2 Performance Evaluation Metrics

The effectiveness of the proposed model was evaluated using standard cybersecurity classification metrics.

Accuracy: Equation 7 measures the overall proportion of correctly classified normal and malicious activities.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

Precision: Precision (equation 8) indicates the percentage of correctly detected attacks among all predicted attacks.

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

Recall: Equation 9 measures the ability of the model to correctly identify actual cyberattacks.

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

F1-Score: Equation 10 provides the harmonic balance between precision and recall performance.

$$F1 = \frac{2\times Precision\times Recall}{Precision+Recall} \quad (10)$$

Detection Rate: Equation 11 evaluates the percentage of malicious attacks successfully identified by the system.

$$Detection\ Rate = \frac{Detected\ Attacks}{Total\ Attacks} \quad (11)$$

False Positive Rate (FPR): False Positive Rate (equation 12) measures the proportion of normal activities incorrectly classified as attacks.

$$FPR = \frac{FP}{FP+TN} \quad (12)$$

4.3 Performance Comparison

The proposed DRL-based anomaly detection model was also compared to other ML and DL techniques like SVM, RF, CNN, and LSTM models.

Table 1 shows a comparison between the traditional ML, DL, and proposed DRL-based anomaly detection model in terms of various cybersecurity metrics. It can be seen that the new DRL-based model exhibited higher values in terms of accuracy, precision, recall, F1 score, and detection rate (98.4%, 98.1%, 97.8%, 97.9%, and 98.2%, respectively), while exhibiting the lowest false positive rate (1.9%).

Table 1: Comparative performance analysis of machine learning and deep reinforcement learning models for phishing and credential stuffing detection in E-learning systems

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Detection Rate (%)	FPR (%)
SVM	90.8	89.7	88.5	89.1	88.9	8.4
Random Forest	93.2	92.5	91.8	92.1	92.6	6.3
CNN	95.1	94.6	94.1	94.3	94.8	4.9
LSTM	96.3	95.7	95.4	95.5	95.9	3.8
Proposed DRL Model	98.4	98.1	97.8	97.9	98.2	1.9

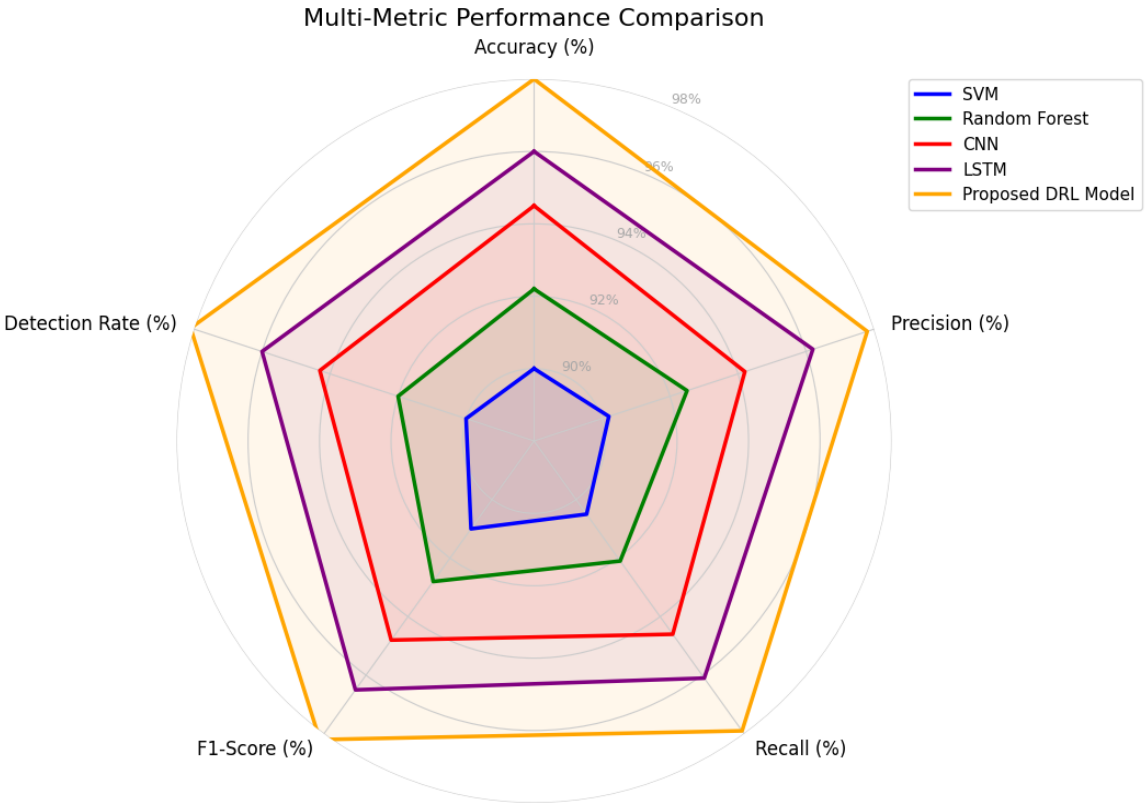


Figure 2: Multi-metric performance comparison of machine learning and deep reinforcement learning models for cyberattack detection

Comparative results for SVM, Random Forest, CNN, LSTM, and the proposed DRL model are depicted in figure 2, from which it can be seen that the proposed DRL model performed best in terms of detection and classification of both phishing and credential stuffing attacks.

It can clearly be observed that the proposed adaptive DRL model performs much better than other baseline models when evaluating using all the different metrics discussed above. The continuous policy

optimization capability of the proposed DRL model allowed it to detect emerging phishing attacks and automated credential stuffing successfully.

Ablation Study

In this section, an ablation study was carried out on the proposed model for determining the individual contributions of its components.

Table 2: Ablation analysis of proposed components

Configuration	Accuracy (%)	F1-Score (%)	Detection Rate (%)
Without Behavioral Features	92.6	91.8	92.1
Without Reward Optimization	93.4	92.9	93.2
Without Sequential Analysis	94.1	93.7	93.9
DRL without Adaptive Learning	95.2	94.6	94.8
Full Proposed Framework	98.4	97.9	98.2

As proved by the ablation test in table 2, adaptive learning, behavior-based feature selection, and sequential anomaly detection all contribute to the better performance of the suggested model. The whole model exhibited the best accuracy for attack detection.

Discussion of Results

However, the proposed anomaly detection approach using DRL was very successful due to its ability to adapt to changes and interact with the environment. Unlike conventional methods like SVM and Random Forest, DRL continuously develops its decision-making policy through environmental feedback and reward optimization. The false positive rate was reduced to 1.9% as compared to the previously existing approaches. Such a reduction in false positives would be highly beneficial in an e-learning setup, as false alarms may affect the learning process negatively. Through the implementation of behavioral analysis, the reinforcement learner was able to detect temporal anomalies, session anomalies, and phishing activities in URLs. Moreover, the experiment shows the efficiency of reinforcement learning along with behavioral analysis in creating scalable, adaptable, and resilient cybersecurity solutions for complex e-learning environments.

5 Conclusion

This research paper discusses the design of an Adaptive Anomaly Detection Algorithm that can help detect any phishing and credential stuffing attacks in real time in e-learning systems based on Deep Reinforcement Learning (DRL). With the growing reliance on distributed Learning Management Systems (LMS), online education services, and online authentication platforms, the threat from cyberattacks to the e-learning infrastructure has substantially grown. Static security models that have been used before have been unable to detect attacks and often result in high false positives, making the system unreliable. The results of the experiment revealed that the developed approach outperformed traditional ML and DL methods, including SVM, Random Forest, CNN, and LSTM in detecting anomalies. Specifically, the proposed approach reached the highest accuracy of 98.4%, precision of 98.1%, recall of 97.8%, F1-score of 97.9%, and relatively low false positives at a level of 1.9%. Through statistical analysis, improvements by 2.1%–7.6% were obtained for anomaly detection accuracy. Moreover, the impact of different aspects of the approach was examined in the ablation study and proved to improve the anomaly detection process by enhancing adaptiveness through adaptive learning, reward

optimization, and behavioral feature extraction. Overall, the results confirm that reinforcement learning can efficiently detect the development of phishing attempts and automated attacks on passwords in real-time conditions in educational organizations. The advantage of adaptive approaches is in the constant updating of policies due to changes in the behavior of attackers. The following topics for further research include leveraging federated learning and XAI methodologies to improve collaboration among threat intelligence models. Other potential topics for research could be blockchain-based authentication systems, zero-trust security frameworks, and lightweight edge-based anomaly detection models for smart education systems.

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